

# Compound Interest Masterclass

Comprehensive Theory, 12 Solved Examples & 25 Practice MCQs with Answer Key

## 1. Introduction & Theoretical Foundation

**Compound Interest (CI)** is the addition of interest to the principal sum of a loan or deposit, or in other words, interest on principal plus interest. It is the result of reinvesting interest, rather than paying it out, so that interest in the next period is then earned on the principal sum plus previously accumulated interest.

Unlike Simple Interest—where interest is calculated solely on the initial principal amount throughout the entire duration—Compound Interest grows exponentially because the base amount increases after every compounding period.

### Core Formula for Amount (A):

$$A = P \left(1 + \frac{R}{100}\right)^n$$

Where:

- $A$  = Total Amount accrued after  $n$  years (Principal + Interest)
- $P$  = Principal amount (initial investment or loan)
- $R$  = Annual rate of interest (in percentage)
- $n$  = Time period (in years)

### Compound Interest (CI) Formula:

$$CI = A - P = P \left[\left(1 + \frac{R}{100}\right)^n - 1\right]$$

## Key Variations & Concepts

- **Compounding more than once a year:** If interest is compounded half-yearly (semi-annually), the rate becomes  $R/2$  and time becomes  $2n$ . If compounded quarterly, rate becomes  $R/4$  and time  $4n$ .
- **Formula for different compounding frequencies:**

$$A = P \left(1 + \frac{R/k}{100}\right)^{n \times k} \text{ (where } k \text{ is the number of times compounded per year).}$$

- **When rates are different for successive years ( $R_1, R_2, R_3$ ):**

$$A = P \left(1 + \frac{R_1}{100}\right) \left(1 + \frac{R_2}{100}\right) \left(1 + \frac{R_3}{100}\right)$$

- **Difference between CI and SI for 2 years:**

$$CI - SI = \frac{P \cdot R^2}{100^2}$$

- **Difference between CI and SI for 3 years:**

$$CI - SI = \frac{P \cdot R^2 (300 + R)}{100^3}$$

## 2. Solved Examples (12 Comprehensive Problems)

### Example 1: Basic Compound Amount & Interest

Find the compound interest on ₹ 5,000 at 10% per annum for 2 years, compounded annually.

**Solution:**

Given:  $P = 5000$ ,  $R = 10\%$ ,  $n = 2$  years.

$$A = 5000 \left(1 + \frac{10}{100}\right)^2 = 5000 \text{ times } (1.1)^2 = 5000 \text{ times } 1.21 = 6050.$$

$$CI = A - P = 6050 - 5000 = \text{₹ } 1,050.$$

### Example 2: Three-Year Compound Interest

Calculate the compound interest on ₹ 10,000 for 3 years at 5% per annum compounded annually.

**Solution:**

$$A = 10000 \left(1 + \frac{5}{100}\right)^3 = 10000 \text{ times } (1.05)^3 = 10000 \text{ times } 1.157625 = 11,576.25.$$

$$CI = 11,576.25 - 10,000 = \text{₹ } 1,576.25.$$

### Example 3: Half-Yearly Compounding

Find the compound interest on ₹ 8,000 at 20% per annum for 1 year, compounded half-yearly.

**Solution:**

Since compounded half-yearly: Rate  $R' = 20/2 = 10\%$  per half-year, Time  $n' = 1 \text{ times } 2 = 2$  half-years.

$$A = 8000 \left(1 + \frac{10}{100}\right)^2 = 8000 \text{ times } 1.21 = 9,680.$$

$$CI = 9,680 - 8,000 = \text{₹ } 1,680.$$

### Example 4: Quarterly Compounding

Find the amount when ₹ 20,000 is invested at 8% per annum compounded quarterly for 9 months.

**Solution:**

Time  $n = 9 \text{ months} = \frac{3}{4} \text{ years}$ . Quarterly rate  $R' = 8/4 = 2\%$  per quarter. Number of quarters  $k = (\frac{3}{4}) \text{ times } 4 = 3$ .

$$A = 20000 \left(1 + \frac{2}{100}\right)^3 = 20000 \text{ times } (1.02)^3 = 20000 \text{ times } 1.061208 = \text{₹ } 21,224.16.$$

### Example 5: Successive Different Rates

Calculate the amount on ₹ 15,000 for 2 years, if the rates of interest for successive years are 10% and 12% respectively.

**Solution:**

$$A = 15000 \left(1 + \frac{10}{100}\right) \left(1 + \frac{12}{100}\right) = 15000 \times 1.10 \times 1.12$$

$$A = 15000 \times 1.232 = \text{₹ } 18,480.$$

### Example 6: Difference Between CI and SI (2 Years)

The difference between compound interest and simple interest on a certain sum of money at 5% per annum for 2 years is ₹ 25. Find the sum.

**Solution:**

$$\text{Formula: } CI - SI = \frac{P \cdot R^2}{100^2} \implies 25 = \frac{P \cdot 5^2}{10000}$$

$$25 = \frac{25P}{10000} \implies P = \frac{25 \times 10000}{25} = \text{₹ } 10,000.$$

### Example 7: Finding Time Period

In how many years will ₹ 4,000 amount to ₹ 5,324 at 10% per annum compound interest?

**Solution:**

$$A = P(1 + R/100)^n \implies 5324 = 4000(1 + 10/100)^n$$

$$\frac{5324}{4000} = (1.1)^n \implies \frac{1331}{1000} = (1.1)^n \implies (1.1)^3 = (1.1)^n \implies \mathbf{n = 3 \text{ years}}$$

### Example 8: Sum Becoming 'K' Times in 'N' Years

A sum of money placed at compound interest doubles itself in 4 years. In how many years will it become 8 times itself?

**Solution:**

If a sum becomes 2 times in 4 years, it will become  $2^3 = 8$  times in  $4 \times 3 = \mathbf{12 \text{ years}}$ .

### Example 9: Principal from Amount after Successive Years

A sum amounts to ₹ 6,655 in 3 years and ₹ 7,320.50 in 4 years at compound interest compounded annually. Find the rate of interest.

#### Solution:

Interest for 1 year on ₹ 6,655 =  $7,320.50 - 6,655 = ₹ 665.50$ .

Rate  $R = \frac{665.50 \times 100}{6655 \times 1} = \mathbf{10\% \text{ ext{ per annum}}}$ .

### Example 10: Population Growth Application

The population of a town is 64,000. If it increases at the rate of 5% per annum, find the population at the end of 3 years.

#### Solution:

$P_3 = 64000 \left(1 + \frac{5}{100}\right)^3 = 64000 \times (1.05)^3 = 64000 \times 1.157625 = \mathbf{74,088}$ .

### Example 11: Depreciation (Negative Compound Growth)

The value of a machine depreciates at the rate of 10% per annum. If its present value is ₹ 50,000, what will be its value after 2 years?

#### Solution:

$V = 50000 \left(1 - \frac{10}{100}\right)^2 = 50000 \times (0.9)^2 = 50000 \times 0.81 = \mathbf{₹ 40,500}$ .

### Example 12: Difference Between CI and SI (3 Years)

Find the difference between compound interest and simple interest on ₹ 8,000 at 10% per annum for 3 years.

#### Solution:

Using formula:  $CI - SI = \frac{P \cdot R^2 (300 + R)}{100^3} = \frac{8000 \times 10^2 \times (300 + 10)}{1000000} = \frac{8000 \times 100 \times 310}{1000000} = \frac{248000000}{1000000} = \mathbf{₹ 248}$ .

## 3. Practice Questions (25 MCQs)

Test your understanding with these 25 multiple-choice questions. Select the best option for each.

**Q1. Find the compound interest on ₹ 4,000 at 5% per annum for 2 years, compounded annually.**

- A) ₹ 400
- B) ₹ 410
- C) ₹ 420
- D) ₹ 450

**Q2. At what rate percent compound interest will ₹ 1,000 amount to ₹ 1,331 in 3 years?**

- A) 5%
- B) 8%
- C) 10%
- D) 12%

**Q3. A sum of money becomes 4 times in 2 years at compound interest. In how many years will it become 16 times?**

- A) 4 years
- B) 6 years
- C) 8 years
- D) 10 years

**Q4. Find the compound interest on ₹ 16,000 for 9 months at 20% per annum compounded quarterly.**

- A) ₹ 2,522
- B) ₹ 2,420
- C) ₹ 2,600
- D) ₹ 2,500

**Q5. The difference between CI and SI on a certain sum for 2 years at 4% per annum is ₹ 1. Find the sum.**

- A) ₹ 500
- B) ₹ 625
- C) ₹ 750
- D) ₹ 1,000

**Q6. What is the compound interest on ₹ 12,500 at 12% per annum for 1 year, compounded half-yearly?**

- A) ₹ 1,536
- B) ₹ 1,545
- C) ₹ 1,500
- D) ₹ 1,600

**Q7. If a sum of ₹ 5,000 amounts to ₹ 7,200 in 2 years at CI, what will it amount to in 4 years?**

- A) ₹ 9,800
- B) ₹ 10,000
- C) ₹ 10,368
- D) ₹ 10,800

**Q8. In what time will ₹ 64,000 amount to ₹ 68,921 at 5% per annum, interest compounded half-yearly?**

- |            |              |
|------------|--------------|
| A) 1 year  | B) 1.5 years |
| C) 2 years | D) 2.5 years |

**Q9. The population of a city is 20,000. It increases by 10% in the first year and by 20% in the second year. Population after 2 years is:**

- |           |           |
|-----------|-----------|
| A) 24,000 | B) 25,000 |
| C) 26,400 | D) 27,200 |

**Q10. A machine worth ₹ 1,00,000 depreciates by 10% every year. Its value after 3 years will be:**

- |             |             |
|-------------|-------------|
| A) ₹ 70,000 | B) ₹ 72,900 |
| C) ₹ 75,000 | D) ₹ 81,000 |

**Q11. Find the compound interest on ₹ 8,000 at 15% per annum for 2 years 4 months, compounded annually.**

- |            |            |
|------------|------------|
| A) ₹ 2,985 | B) ₹ 3,010 |
| C) ₹ 3,105 | D) ₹ 3,250 |

**Q12. The compound interest on a certain sum for 2 years is ₹ 410 and simple interest is ₹ 400. Find the rate of interest.**

- |       |        |
|-------|--------|
| A) 2% | B) 4%  |
| C) 5% | D) 10% |

**Q13. If the interest is compounded half-yearly, principal is ₹ 10,000 and rate is 10% p.a., amount after 1 year is:**

- |             |             |
|-------------|-------------|
| A) ₹ 11,000 | B) ₹ 11,025 |
| C) ₹ 11,050 | D) ₹ 11,250 |

**Q14. A sum amounts to ₹ 2,400 in 2 years and ₹ 3,000 in 3 years under CI. Find the annual rate of interest.**

- |        |        |
|--------|--------|
| A) 15% | B) 20% |
| C) 25% | D) 30% |

**Q15. The difference between CI and SI on a sum for 3 years at 10% per annum is ₹ 31. Find the sum.**

- |            |            |
|------------|------------|
| A) ₹ 800   | B) ₹ 900   |
| C) ₹ 1,000 | D) ₹ 1,200 |

**Q16. At what rate percent per annum will ₹ 2,304 amount to ₹ 2,500 in 2 years at compound interest?**

- |       |       |
|-------|-------|
| A) 4% | B) 5% |
| C) 6% | D) 8% |

**Q17. A sum of ₹ 12,000 deposited at compound interest becomes double after 5 years. After 20 years, it will become:**

- |               |               |
|---------------|---------------|
| A) ₹ 96,000   | B) ₹ 1,44,000 |
| C) ₹ 1,92,000 | D) ₹ 2,40,000 |

**Q18. What principal will amount to ₹ 5,832 in 2 years at 8% per annum compound interest?**

- |            |            |
|------------|------------|
| A) ₹ 4,500 | B) ₹ 4,800 |
| C) ₹ 5,000 | D) ₹ 5,200 |

**Q19. Find the compound interest on ₹ 10,000 for 1 year at 10% per annum compounded quarterly.**

- |               |               |
|---------------|---------------|
| A) ₹ 1,038.13 | B) ₹ 1,050.00 |
| C) ₹ 1,100.00 | D) ₹ 1,150.50 |

**Q20. Divide ₹ 3,903 between A and B such that A's share after 4 years is equal to B's share after 6 years at 4% per annum CI.**

A) ₹ 2,028, ₹ 1,875

B) ₹ 2,000, ₹ 1,903

C) ₹ 2,100, ₹ 1,803

D) ₹ 2,050, ₹ 1,853

**Q21. The simple interest on a sum for 2 years is ₹ 160 and compound interest is ₹ 164. Find the principal sum.**

A) ₹ 1,000

B) ₹ 1,200

C) ₹ 1,500

D) ₹ 1,600

**Q22. A man borrows ₹ 10,000 at 10% compound interest per annum. At the end of each year he pays back ₹ 3,000. How much amount is left unpaid after 3 years?**

A) ₹ 3,210

B) ₹ 3,310

C) ₹ 3,450

D) ₹ 3,600

**Q23. If the amount is 2.25 times of the sum after 2 years at compound interest, the rate of interest per annum is:**

A) 10%

B) 15%

C) 20%

D) 50%

**Q24. A sum of ₹ 3,000 yields ₹ 993 as compound interest in 3 years. Find the rate of interest.**

A) 5%

B) 8%

C) 10%

D) 12%

**Q25. If the annual rate of interest increases from 10% to 12.5%, a man's yearly income increases by ₹ 1,250. Find the principal sum.**

A) ₹ 25,000

B) ₹ 40,000

C) ₹ 50,000

D) ₹ 60,000

### Answer Key (Practice Questions 1 to 25)

| Q.No | Ans | Q.No | Ans | Q.No | Ans | Q.No | Ans | Q.No | Ans |
|------|-----|------|-----|------|-----|------|-----|------|-----|
|------|-----|------|-----|------|-----|------|-----|------|-----|

|   |   |    |   |    |   |    |   |    |   |
|---|---|----|---|----|---|----|---|----|---|
| 1 | B | 6  | A | 11 | A | 16 | B | 21 | D |
| 2 | C | 7  | C | 12 | D | 17 | C | 22 | A |
| 3 | B | 8  | A | 13 | B | 18 | C | 23 | D |
| 4 | A | 9  | C | 14 | C | 19 | A | 24 | C |
| 5 | B | 10 | B | 15 | C | 20 | A | 25 | C |